

RAINNE BUSQUE

Phoenix, AZ | rbusque@asu.edu | <https://tinyurl.com/3aupbmb6> | <https://github.com/rainEBB13>

EDUCATION

Arizona State University

Tempe, AZ

Bachelor of Science in Data Science (Mathematics Track)

Aug. 2024 – May 2029

- Dean's Scholarship Recipient
- Relevant Coursework: Linear Algebra, Calculus II (MAT266), Data Structures, Introduction to Machine Learning
- Organizations: Devil Data Club, Google Developer Club, Philippine American Student Association (Culture Chair)

EXPERIENCE

Undergraduate Research Assistant

May 2026 – Jul. 2026

ASU OURS — Dr. Jimmie Adriaola (NSF ASCEND Fellow)

Remote

- Contributed to *Learning Mathematics Through the Machine*, a research project using statistical learning to rediscover classical mathematical formulas from numerical data
- Implemented and analyzed Sparse Identification of Nonlinear Dynamics (SINDy) and sparse regression pipelines in Python and MATLAB
- Applied symbolic regression and numerical methods including Euler's method to recover differential equation structure from data
- Developed understanding of linear algebra foundations underpinning modern ML algorithms through active research application

Hackathon Participant — NutriWell

Aug. 2025

Project Humanities Hackathon

- Built a web app connecting food-insecure individuals to nearby donation centers and donors as part of a team
- Focused on accessible UX design and real-world social impact; developed under time-constrained hackathon conditions

DataFest Competitor

Spring 2026

ASU Devil Data Club

- Analyzed a complex healthcare infrastructure dataset under time pressure as part of a competitive data analytics team
- Performed Exploratory Data Analysis (EDA) to uncover hidden patterns within the dataset.

Fitness Team Member

2026 – Present

Planet Fitness

Tempe, AZ

- Delivered consistent member service while managing time effectively alongside a full academic and research schedule

PROJECTS

Philippine Housing Prices Prediction | *Python, scikit-learn, pandas, NumPy*

2026

- Built a supervised regression model to predict residential housing prices in the Philippines using real estate data
- Performed data cleaning, feature engineering, and exploratory analysis to identify key price drivers such as location, size, and property type
- Trained and evaluated multiple models including linear regression, decision trees, and ensemble methods; tuned hyperparameters to optimize predictive accuracy

Computer Vision Pipeline | *Python, YOLOv8, OpenCV*

2025

- Designed and deployed an object detection pipeline using YOLOv8 for real-time inference; implemented custom preprocessing with OpenCV

Terminal Spinning Donut Renderer | *Python, NumPy*

2025

- Re-implemented the classic ASCII donut renderer using NumPy vectorized math for 3D rotation projection in the terminal

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, L^AT_EX

ML & Data Science: scikit-learn, pandas, NumPy, Matplotlib, Plotly, YOLOv8, OpenCV

Methods: SINDy, Sparse Regression, Symbolic Regression, Supervised Learning, Decision Trees, Neural Networks, Euler's Method

Tools & Platforms: Git/GitHub, Jupyter Notebook, VS Code, Vim, tmux, Arduino, Linux/macOS

LEADERSHIP & ACTIVITIES

Culture Chair

2026 – 2027

Philippine American Student Association (PASA), Arizona State University

- Elected to lead cultural programming and community engagement initiatives for the organization
- Organized events celebrating Filipino-American heritage and fostering a sense of belonging on campus